James Kirika

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INTRO

Electrical and Electronics Engineering graduate with three-plus years of freelance programming experience writing MQL4/5 software for MetaTrader clients. I'm a fast, self-directed learner; I've picked up C++, C#, Python, and Node.js on the job and through personal projects, working from documentation and hands-on practice rather than formal training. Much of my recent work already brushes up against security fundamentals: building encrypted WebSocket (WSS) connections, managing SSL certificates, and implementing token-based authentication for live data bridges between MetaTrader 5 and web servers. My final-year engineering project was an anti-tampering smart meter system built to detect unauthorized access. I'm currently pursuing the Cyber Shujaa Cloud and Network Security track to turn that instinct into formal cybersecurity skills, bringing the same curiosity and quick-learning approach that's gotten me this far.

WORK EXPERIENCE

Software Engineer (Freelance) — MQL5 Freelance **Apr 2023 – Present**

Roles

- Write MQL4/MQL5 programs for trading platforms, including custom indicators, scripts, and Expert Advisors (EAs).
- Convert trading logic from TradingView's Pine Script into MQL4/5 for clients migrating platforms.
- Build C++ and C# DLL bridges that stream live MT5 data to web dashboards and Excel over secure WebSocket (WSS) connections, using SSL certificates and token-based authentication.
- Debug and troubleshoot code and provide ongoing technical support as client needs evolve.

Key Achievements

- Developed and deployed custom DLLs, trading indicators, scripts, and Expert Advisors that meaningfully improved trading efficiency and execution speed for clients.
- Delivered secure real-time bridges authenticated with SSL certificates and tokens.
- Consistently praised by clients for the reliability and usability of delivered work, and for meeting tight deadlines.

Technician — LED Power Technology **Jan 2022 – Dec 2022**

Roles

- Installed and maintained CCTV and IP camera systems: routed structured cabling, mounted cameras, and configured network switches to get footage onto the local network.
- Conduit work and electrical installation (wiring and fitting).
- Installation and maintenance of single-phase and three-phase motors and HVAC systems, and genset servicing.

Key Achievements

- Proved to be a quick learner, critical thinker, time-conscious, responsible, and a great team player.

Attachee — KPLC **Jan 2021 – Apr 2021**

Roles

- Low voltage transformer wiring and cable connection on meter bus bars.
- Installation of large power meters and current transformers.
- Power disconnection and reconnection, cable stripping and crimping, and meter housing assembly.
- Pole and wall mounting of smart meters, and meter reading.

Key Achievements

- Proved to be a disciplined intern who upheld good ethical values both socially and duty-wise.

TECHNICAL PROJECTS

Secure MT5-to-Web Bridge

Designed and built DLLs (C++/C#) that stream live MetaTrader 5 trading data to browser dashboards and Excel in real time. The pipeline runs from MT5 Expert Advisor, through the DLL, over WebSocket, into a Node.js server, and out to the browser. Implemented WSS with both self-signed and CA-trusted SSL certificates depending on deployment target, added token-based client authentication, and deployed via Cloudflare Tunnel and Render for always-on hosting. Code: github.com/Sir-kirika/mt5bridge.

Web Application Security Lab (Flask / SQLite)
Built a Flask and SQLite web application, then found and exploited a SQL injection vulnerability in it and remediated it with parameterized queries. Also fixed an API that changed data through GET requests and removed the server version banner to reduce information disclosure. Write-up on my portfolio.

Notepad Calculator (Python / Kivy)
Built a desktop note-taking app for personal use that combines free-form notes with inline arithmetic: supports variable assignment and reuse within a note, light/dark themes, and undo/redo, designed and coded solo from the UI down to the expression-parsing logic.

Quantitative FX Research
Self-directed research applying Hidden Markov Models to detect volatility regimes across FX pairs, and Kuramoto synchronization theory (borrowed from physics) to study coordinated trader behavior around gold and major USD pairs, including an 8-year walk-forward backtesting and optimization pipeline built in Python.

EDUCATION, CERTIFICATIONS & TRAINING

Cyber Shujaa — Cloud and Network Security Track **In progress, Sep 2026 intake**

- Structured cybersecurity training covering networking fundamentals, cloud security, and hands-on labs (Cisco Packet Tracer, Wireshark, VLAN hardening, site-to-site VPN), building formally on existing hands-on experience with WSS/SSL and secure data pipelines.

Bachelor of Technology in Electrical and Electronics Engineering **Jan 2017 – Dec 2021**
Murang'a University of Technology • Second Class Honours, Upper Division

KEY SKILLS & COMPETENCIES

Programming & Software

- MQL4/MQL5 (Indicators, EAs, Scripts); Pine Script to MQL4/5 conversion
- C++ / C# (DLL development)
- Python (Kivy, Flask, data analysis); Node.js
- WebSockets, SSL/TLS, token-based authentication, secure data pipelines
- Networking and security fundamentals: Cisco Packet Tracer, Wireshark, VLANs, IPsec VPN; SQL injection testing and remediation
- Computer-Aided Design (Archicad, AutoCAD, Proteus)
- Git / GitHub
- Office tools: Word, Excel, PowerPoint

Electrical & Technical

- Circuit Design and Implementation
- Electrical Installation and Assembly
- High Voltage Engineering & Power Line Construction
- Three-Phase Networks (Electricity)
- CCTV / Analogue and IP Camera Installation (cabling, mounting, switch configuration)

Soft Skills

- Quick, self-directed learning; critical thinking and problem solving
- Teamwork and collaboration; client communication and technical support
- Time management and meeting tight deadlines
- Responsibility, discipline, ethical conduct, adaptability, and attention to detail

Other Skills

- First Aid and firefighting skills

ASSOCIATIONS

Member, MUT Society of Engineering Technology

HOBBIES

Cycling, programming, and travelling.

REFEREES

Available on request.